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AMAZON AIR FRYERS

SJ & SYD

{ A DEMAND ESTIMATION PROJECT }



(04)



(05)



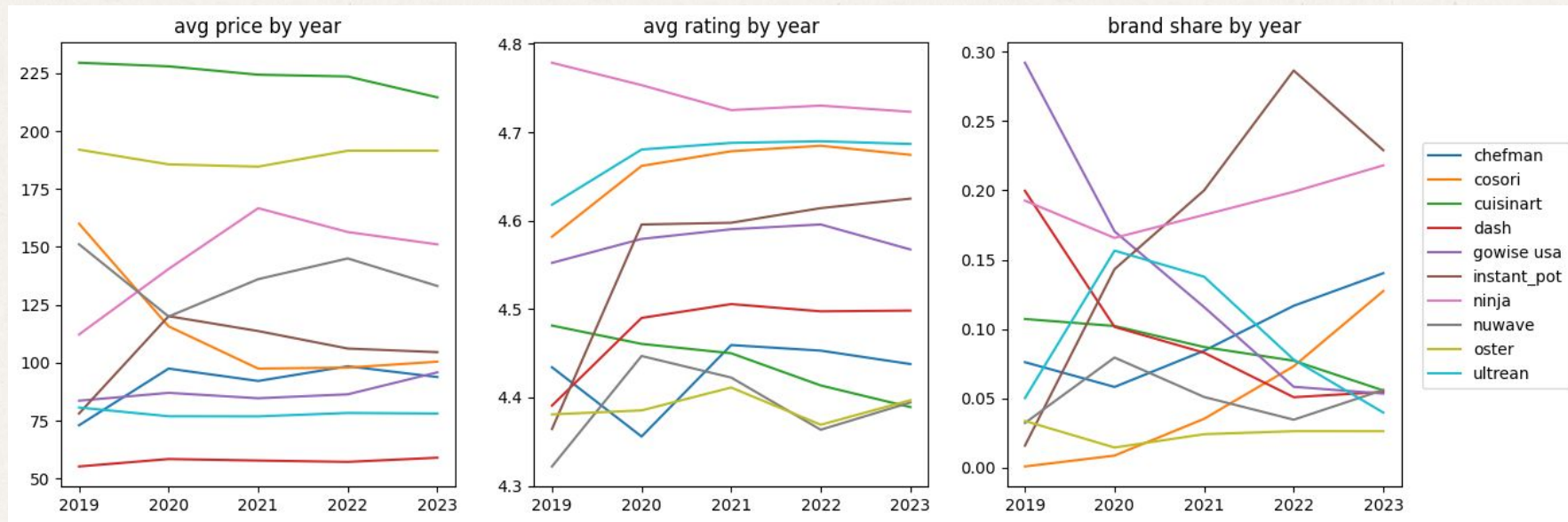
(06)

The Air Fryer Market

- Market doubled from ~15K to ~31K purchases by 2021, then pulled back, likely a pandemic bump
- Ninja and Instant Pot now dominate with ~44% combined share; GoWise USA fell from 29% to 5%
- Cosori was the fastest riser, going from near-zero to ~13%
- Prices held flat at ~\$122-\$124 despite volume swings; ratings are uniformly high (4.3-4.8)
- Compact basket style dominates at 95%+; dual-basket and window formats are just emerging



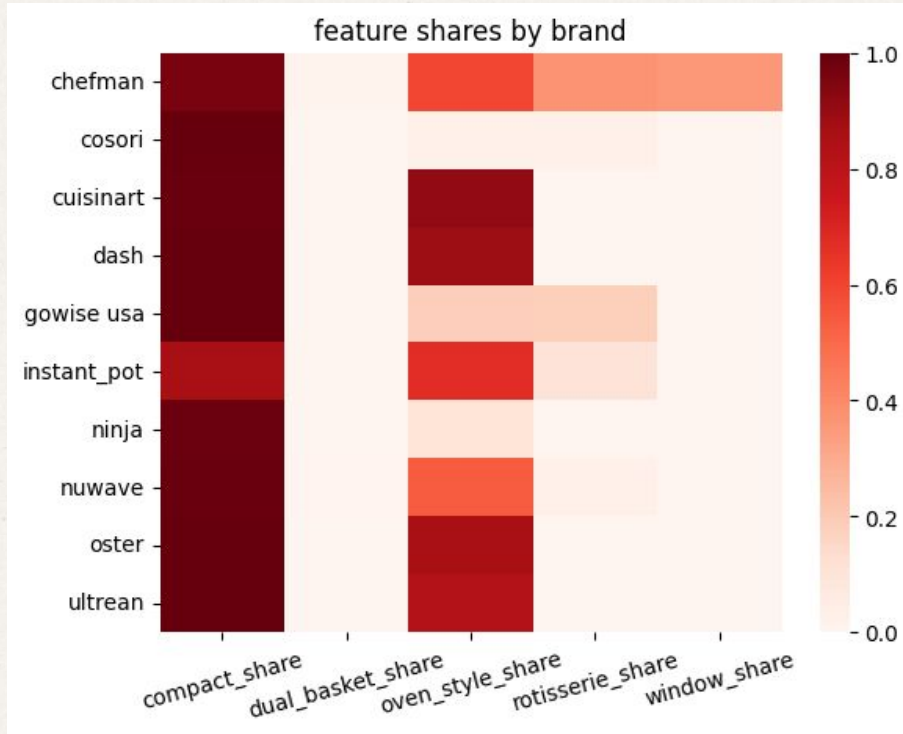
OVERALL MARKET



Average Price by Brand - Stable throughout. Cuisinart and Oster are premium (~\$190-\$230); Dash stays budget (~\$57). Average Rating by Brand - All brands rate 4.3-4.8 with little change over time. Effectively no differentiation. Brand Market Share Over Time - Most volatile metric. GoWise USA collapsed (30%→5%); Instant Pot surged to market leader. Ninja and Instant Pot dominate by 2023.

(MADE USING MATPLOTLIB
AND SEABORN)

OVERALL MARKET

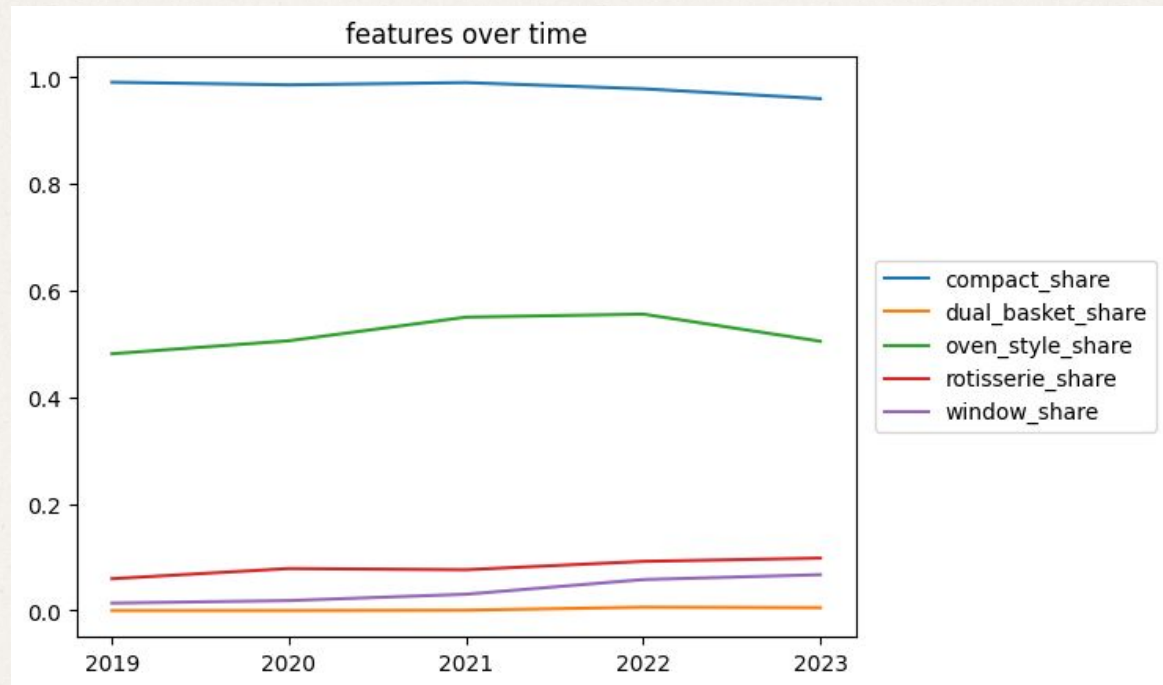


- Compact design is near-universal across all brands (95%+)
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- Cuisinart, Dash, and Oster stand out for oven-style offerings
- GoWise USA and NuWave lead on rotisserie
- Dual-basket is virtually absent, only just emerging post-2021

Demand Model & Price Coefficient

- Our model predicts market share based on price, rating, product features, brand, and year.
- It uses multinomial logistic regression, estimated by running linear regression on log market share, where each coefficient captures the proportional change in share associated with a one-unit change in that variable.
- For example, the price coefficient is -0.0377 : a \$1 increase in price is associated with a $\sim 3.77\%$ decrease in market share, confirming that demand falls as price rises. Negative means higher price leads to lower share, as basic economics predicts. A positive sign would be a red flag.

FEATURES IN THE MARKET



- Compact (98.1%): The dominant form factor across the entire market
- Oven-style (52.0%): Present in over half of all models, the clear second most common feature
- Rotisserie (8.1%): A niche add-on, small and stable over time
- Window (3.8%): Rare but slowly growing
- Dual-basket (0.3%): Nearly nonexistent, only beginning to emerge after 2021

MOST IMPORTANT FEATURES

Window

Rare (3.8%) but the strongest positive signal. Brands offering a window are associated with meaningfully higher market share.

Compact

The market standard at 98.1%. Its positive coefficient confirms that the dominant form factor also drives demand.

Oven-style

Present in half the market. Positively associated with share, suggesting consumers value the versatility.

Rotisserie

A niche feature at 8.1%. Negative coefficient, not associated with higher demand despite its specialization.

Dual-basket

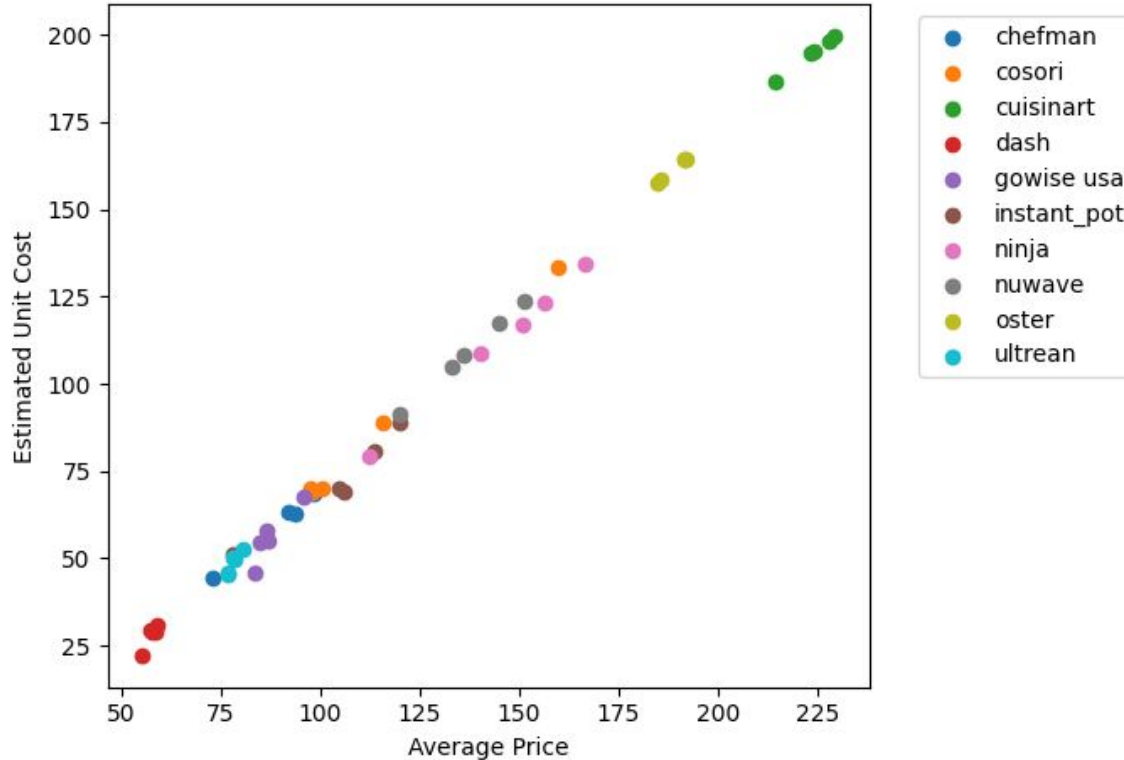
Nearly absent at 0.3%. Negative coefficient, though too rare to draw strong conclusions.



UNIT COSTS

{ STORIES TOLD
THROUGH FOOD }

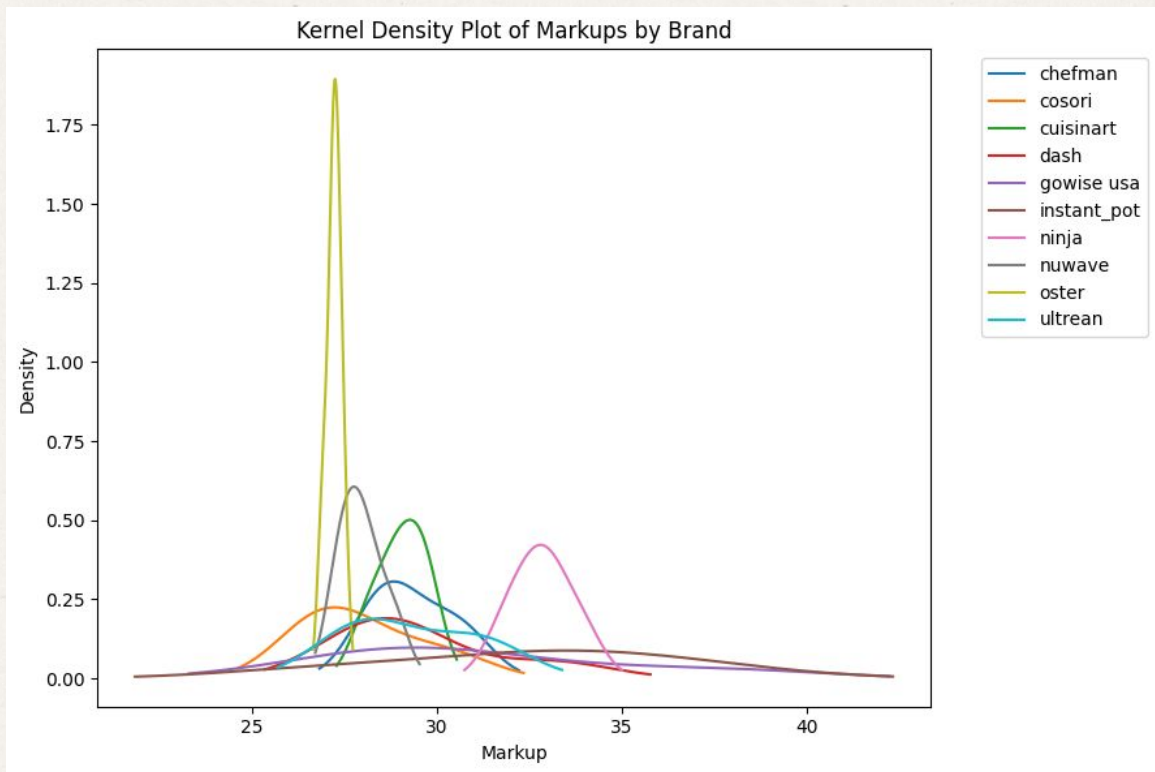
Price vs. Unit Cost



Higher-priced brands have higher estimated unit costs. Cuisinart and Oster sit at the top; Dash and Ultrean at the bottom. The relationship is fairly linear across brands.

Markups

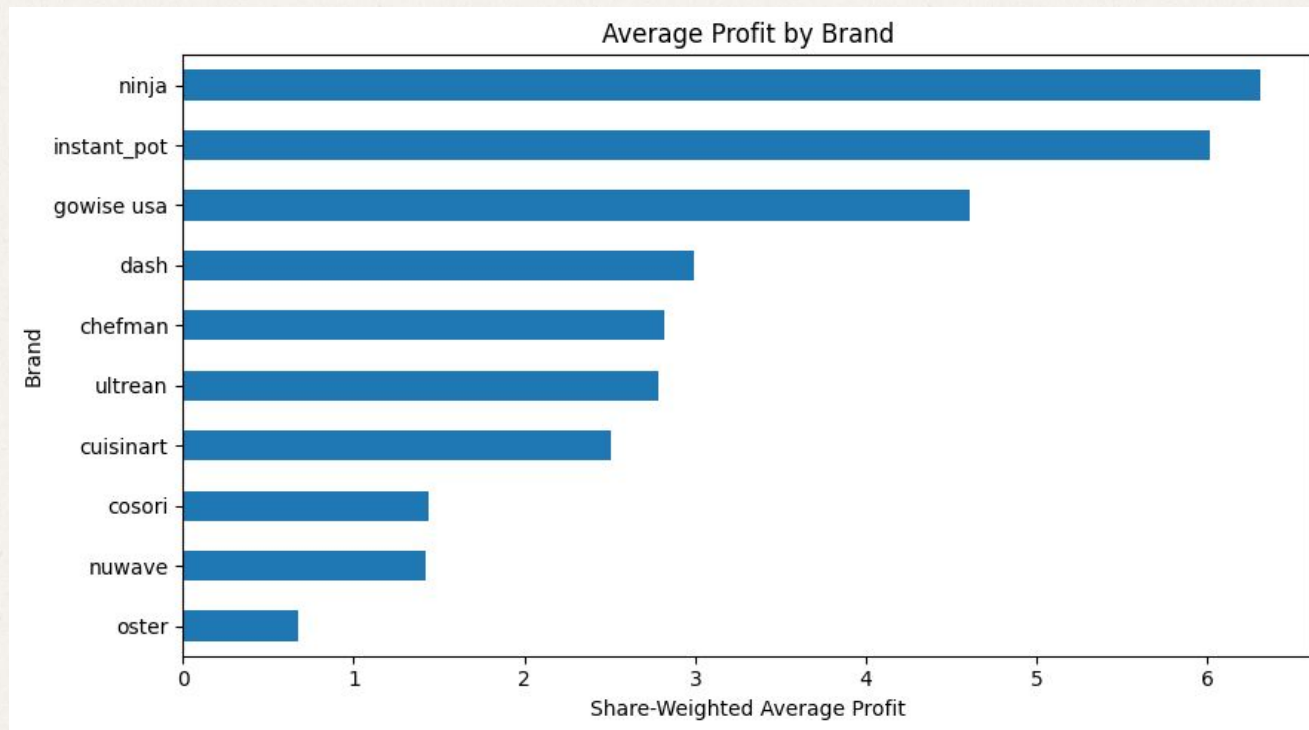
{ STORIES TOLD
THROUGH FOOD }



Most brand-year markups cluster in a relatively narrow range, with a long tail toward higher values. A small number of brand-year observations capture disproportionately high margins.

Shared-weight profits

{ STORIES TOLD
THROUGH FOOD }



Ninja and Instant Pot lead on share-weighted profit, combining strong margins with large market share. Brands with high unit costs like Cuisinart do not necessarily convert that into high profit.

CONCLUSION

Ninja and Instant Pot are the clear leaders, combining large share with strong markups and the highest profits. Cuisinart and Oster hold a premium niche but modest share limits their profitability. Cosori is the standout grower, gaining share rapidly while staying competitively priced. GoWise USA tells the opposite story, collapsing from 30% to 5% share over five years.



THANK YOU